

Sample Research Paper

Abstract: This paper introduces a retrieval-augmented generation system that uses NVIDIA NIM embeddings and ChromaDB for indexing.

Section 1: Introduction

RAG systems combine dense retrieval with large language models to answer questions grounded in a private corpus. We use llama-3.1-70b for answers and nv-embedqa-e5-v5 for embeddings. Retrieval is cosine over ChromaDB.

Section 2: Method

PDFs are parsed with PyMuPDF into per-page text. We chunk by section boundaries and embed each chunk. Multimodal extraction yields figures, tables, equations and captions linked to their source page.

Section 3: Results

The system retrieves top-k chunks and the VLM describes figures and tables so they can be matched by semantic similarity to user questions.